3 Memory Mapping

Introduction

Memory is an integral part of a microcomputer system. It is used to store instructions and data in binary format for the microprocessor. Memory can be of two types, that is, prime (or main) and storage memory. Read/write memory (R/W M) and read only memory (ROM) are the examples of main or prime memory and magnetic tapes or hard discs are the examples of storage or secondary memory.

The R/W memory consists of registers, which are made of a group of flip-flops that store bits of information. These flip-flops are called memory cells. The number of bits stored in a register is called a memory word. In a memory chip, all registers are arranged in sequence and identified by binary numbers called memory addresses. To communicate with memory, the Memory Protection Unit (MPU) should be able to do the following.

- Select the chip
- Identify the register
- Read from or write into the register

The MPU uses its address bus to send the address of a memory register and uses the data bus and control lines to read from or write into that register.

Building of Memories

A flip-flop or latch is a basic element of memory. Tri-state buffers, as shown in the following figures, can be used on the input and output side of the latch to avoid uninternational change in the input and control the availability of the output.

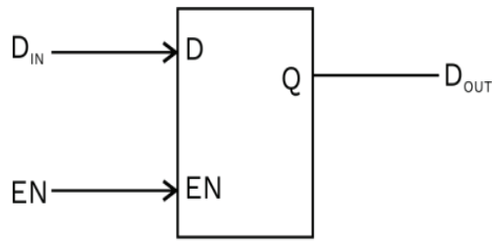


Fig. 3.1 Latch as Memory Cell

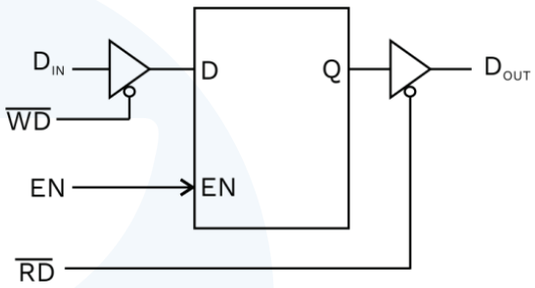


Fig. 3.2 Latch with Tri-state Buffers

Input data can be written into the latch by enabling the input buffers using active low $\overline{WR}$ signal and can be read from it by enabling the output buffers using active low $\overline{RD}$ signal. This latch, with read and write, enables signals, which can store one binary bit called memory cell.

Fig. 3.3 shows an 8 bit register consisting of eight such cells or latches grouped together.

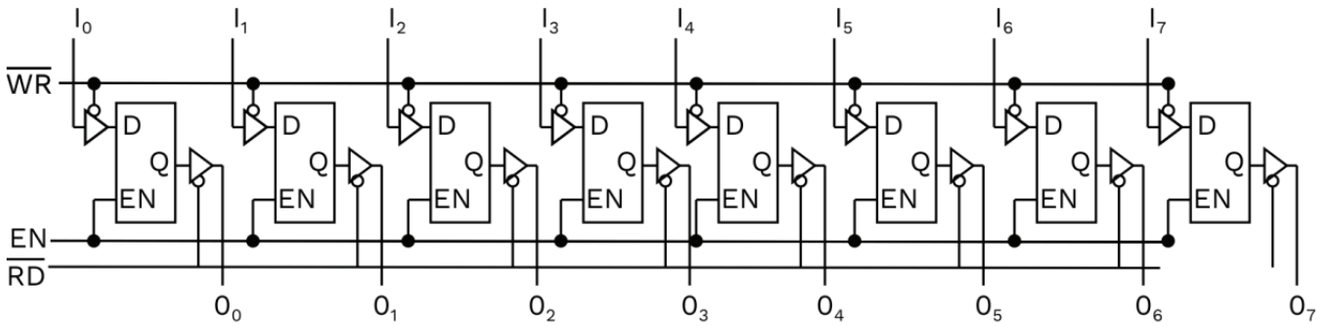


Fig. 3.3 Eight Bit Register

The figure given below shows simplified block diagrams of an 8 bit register with input and output buffers.

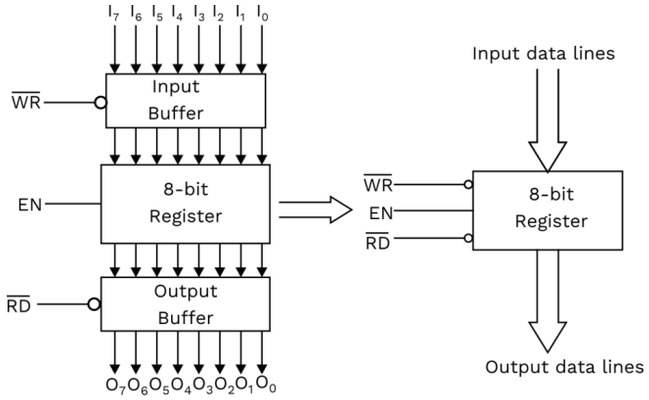


Fig. 3.4 Simplified Block Diagram of an 8 Bit Register

In Fig. 3.5, four registers with eight cells (or an 8 bit memory word) are arranged in a sequence. It is an example of 4 byte memory. A two to four decoder can be used for enabling or giving clock pulses to register.

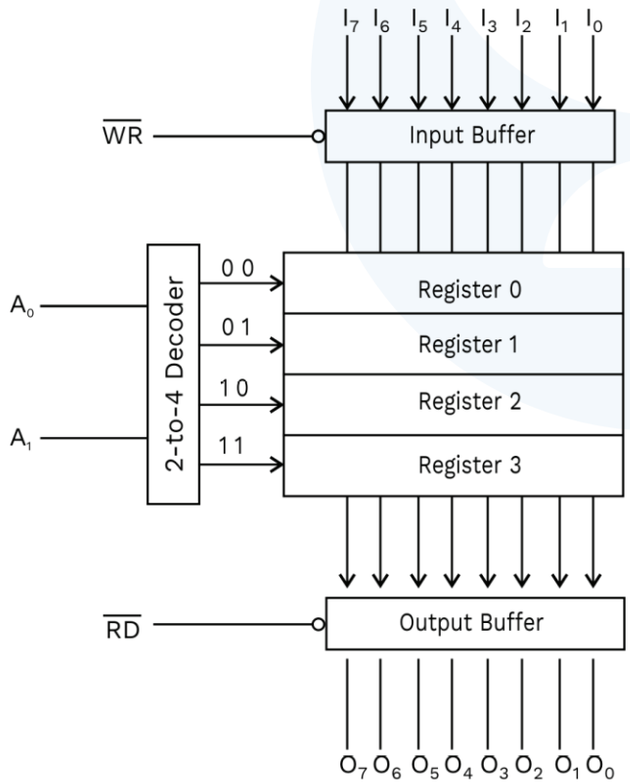


Fig. 3.5 Four Byte Memory

Two chips with 4 bit registers can also be connected to make an 8 bit word memory size as shown in Fig. 3.5.

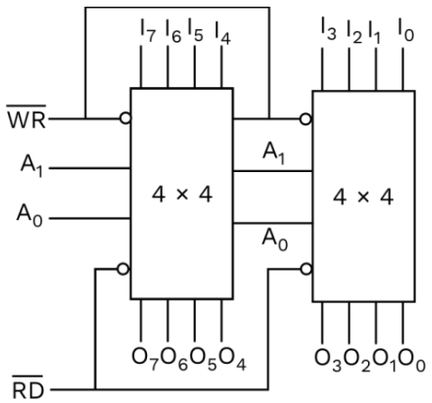


Fig. 3.6 Four Byte Memory using two 4 Bit Memories

The size of the memory can be further expanded by taking eight registers on one chip. Three address lines can be used to enable the register in memory chips.

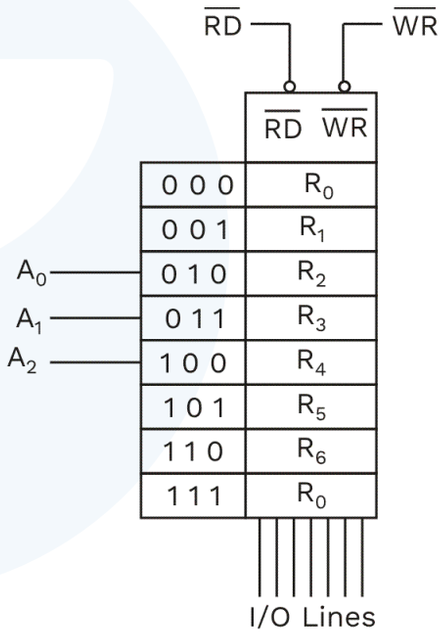


Fig. 3.7 Eight Byte Memory with Three Address Lines

More than one chip can also be used to form larger sizes of memory. For example, two chips with four 8 bit registers each can be used to construct a memory with either register. In such a case, we need three address lines, out of which one line is used to select between the two chips and the remaining two lines are used to select a register in the chip.

Fig. 3.5 shows two memory chips, with an additional signal called chip select (CS), and A_2, with an inverter, are used to select between the chips.

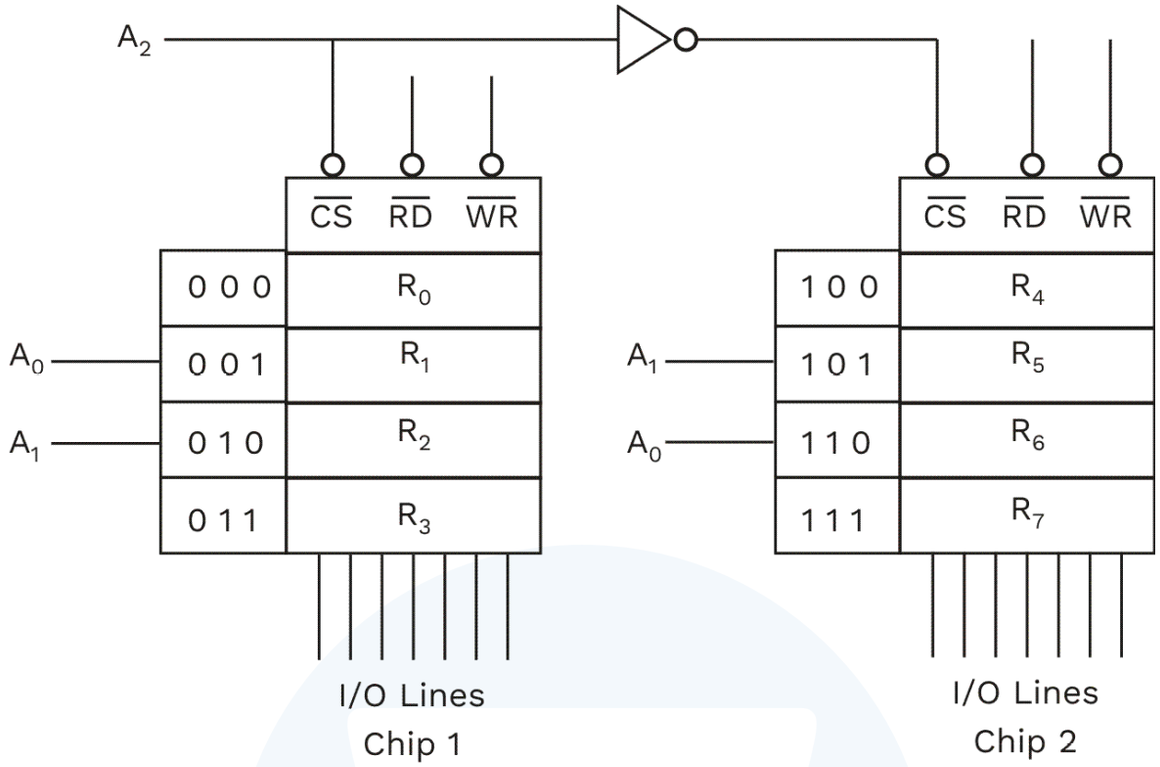


Fig. 3.8 Two Memory Chips with Chip Select

When we have four address lines and two memory chips with four registers, each of the two lower order address lines can be used to enable the register of the chip and two higher order lines can be used to generate chip select logic as shown in Fig. 3.9.

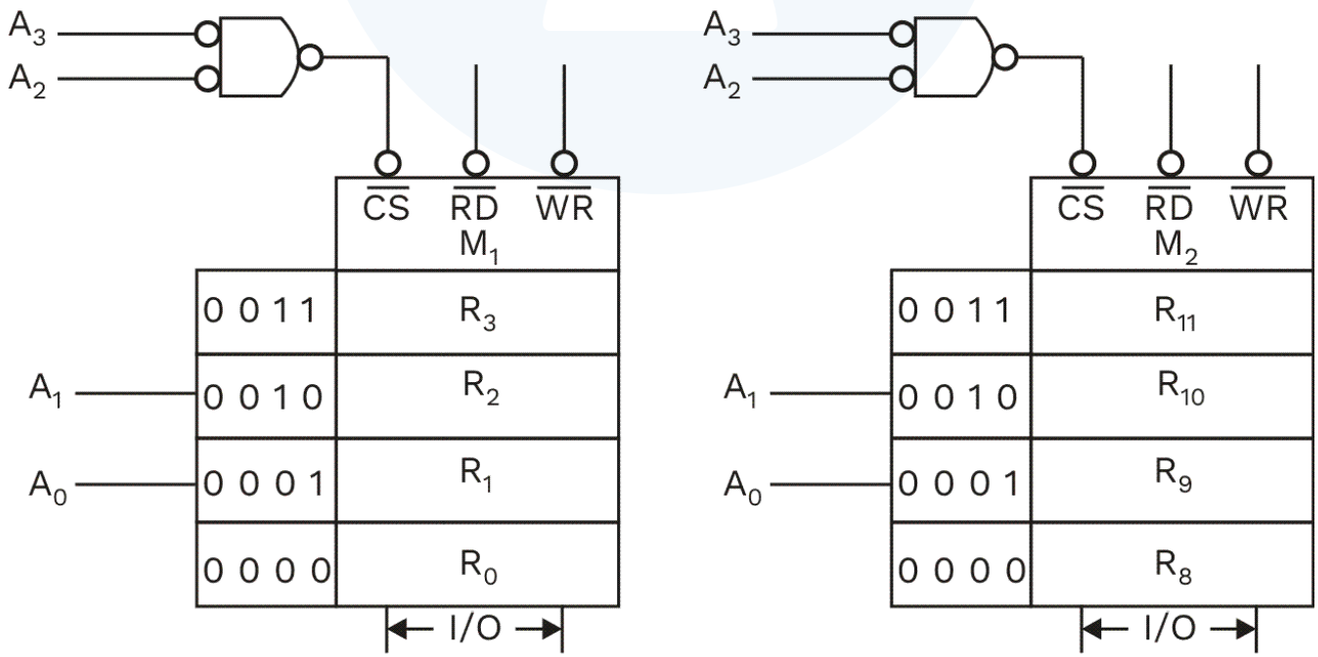


Fig. 3.9 Address Eight Registers with Four Address Lines

Practice Exercise

Level I

1. The address bus of Intel is 16 bit wide, and hence, the memory which can be accessed by this address bus is: **[ESE-EE-92]**
(A) 2 k bytes
(B) 4 k bytes
(C) 16 k bytes
(D) 64 k bytes

2. The number of bits needed to address 4K memory is: **[ESE-EE-97]**
(A) 6
(B) 8
(C) 12
(D) 16

3. Three memory chips are of size 1 KB, 2 KB, and 4 KB. Their address bus is 10 bits. What are the data bus sizes of the chips? **[ESE-E&T-07]**
(A) 8 bits, 16 bits, and 24 bits, respectively
(B) 8 bits, 16 bits, and 32 bits, respectively
(C) 8 bits, 16 bits, and 64 bits, respectively
(D) 8 bits, 16 bits, and 128 bits, respectively

4. Memory chips of four different sizes as given below are available.
I. 32K×4
II. 32K×16
III. 8K×8
IV. 16K×4
All memory chips as mentioned in the above list are read/write memory. What minimal combination of chips or chip alone can map the full address space of an 8085 microprocessor? **[ESE-EE-05]**
(A) I and II
(B) I only
(C) II only
(D) IV only

5. What is the number of memories required of size 16K × 4 to design a memory of size 64K × 8? **[ESE-EE-09]**
(A) 2
(B) 4
(C) 6
(D) 8

6. To address the memory, 14 bits are used. Then, what is the address of the last memory location? **[ESE-EE-09]**
(A) 16382
(B) 16383
(C) 16384
(D) 16385

7. A 3 × 8 decoder with two enable inputs is to be used to address eight blocks of memory. What will be the size of each memory block when addressed from a 16 bit with two MSBs used to enable the decoder? **[ESE-E&T,09]**
(A) 2 K
(B) 4 K
(C) 16 K
(D) 64 K

8. A 16 bit memory address register can address the memory locations of: **[ESE-EE-10]**
(A) 16k
(B) 32k
(C) 64k
(D) 128k

9. The number of memory chips of size 1K × 4 bits required to build a memory bank of size 16K × 8 bits is: **[ESE-03]**
(A) 64
(B) 32
(C) 16
(D) 8

10. In a memory map, the lowest address of an 8K byte RAM is 1000 H. What is the highest address? **[ESE-06]**
(A) 8192 H
(B) 2FFF H
(C) 7FFF H
(D) 3000 H

11. On a certain memory chip is written 4K × 8; it implies that it has the data input lines, data output lines, and capacity, respectively as: **[ESE-02]**

(A) 8,8,4000
(B) 8,8,4096
(C) 12,12,4096
(D) 8,12,4096

12. Eight memory chips of size 64 × 8 bits have their address buses connected together. What will be the size of the resultant memory? **[IAS-09]**
(A) 64 × 64 bits
(B) 128 × 32 bits
(C) 256 × 16 bits
(D) 512 × 8 bits

13. An 8 bit micro controller has an external RAM with a memory map from 8000H to 9FFFH. The number of bytes this RAM can store is: **[ISRO-09]**
(A) 8193
(B) 8192
(C) 8191
(D) 8000

14. A memory system of size 16K bytes is required to be designed using memory chips, which have 12 address lines and four data lines each. Then, the number of such chips required to design the memory system is: **[ISRO-09]**
(A) 2
(B) 4
(C) 8
(D) 16

15. The number of address lines required to address 8 k bytes of memory is: **[BSNL-JTO-TC-09]**
(A) 13
(B) 14
(C) 15
(D) 16

Level II

1. A memory system has a total of eight memory chips, each with 12 address lines and four data lines. The total size of the memory system is: **[GATE-EE-03]**
(A) 16k bytes
(B) 32K bytes
(C) 48K bytes
(D) 64K bytes

2. An 8 bit micro controller has an external RAM in the memory map from 8000H to 9FFFH. The number of bytes this RAM can store is: **[GATE-EE-03]**
(A) 8193
(B) 8191
(C) 8192
(D) 8000

3. An I/O peripheral device, shown in Fig. (b), is to be interfaced with an 8085 microprocessor. To select the I/O device in the I/O address range D4 H–D7 H, its chip-select $(\overline{CS})$ should be connected to the output of the decoder, shown in Fig. (a). **[GATE-EC-06]**

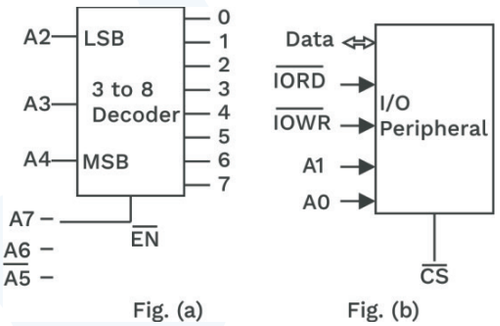


Fig. (a) Fig. (b)

Fig. 3.10

(A) Output 7
(B) Output 5
(C) Output 2
(D) Output 0

4. An 8085 microprocessor based system uses a 4K×8 bit RAM whose starting address is AA00 H. The address of the last byte in this RAM is: **[GATE-EC-01]**
(A) 0FFF H
(B) 1000 H
(C) B9FF
(D) BA00 H

5. An 8 bit DAC is interfaced with a microprocessor having 16 address lines (A_0...A_{13}) as shown in the adjoining figure. A possible valid address for this DAC is: **[GATE-IN-10]**

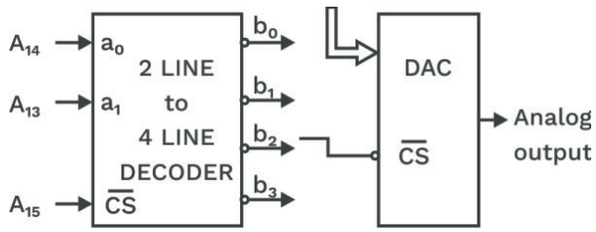


Fig. 3.11

(A) 3000_H
(B) $4FFF_H$
(C) $AFFF_H$
(D) $C000_H$

6. **Assertion (A):** Tri-state logic is used for bus oriented systems.

Reason (R): Tri-state logic has three output states: 0, 1, and indeterminate. **[ESE-E&T-91]**

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

7. For 16 bit address bus, if an 8K RAM chip is selected when A_{13}, A_{14}, and A_{15} address bits are all one, then what is the range of the memory address? **[ESE-EE-09]**

(A) E000H–EFFFH
(B) E000H–FFFFH
(C) F000H–FFFFH
(D) F00H–FEEEH

8.

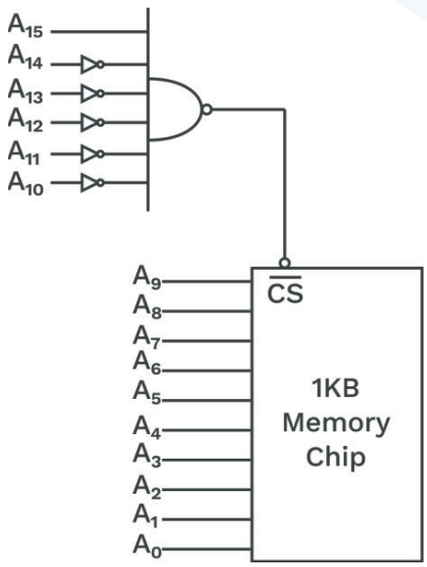


Fig. 3.12

Consider a memory chip with 1024 bytes of storage connected to 8085 chip address lines (or any microprocessor with 16 address lines) as above. What is the range of memory address? **[ESE-E&T-10]**

(A) 000 H to 03 FFH
(B) 8000 H to 8 FFH
(C) F000 H to F3FFH
(D) 0000 H to FFFFH

9. **Assertion (A):** The port address in IN/OUT instruction is 8 bit.

Reason (R): The address byte is duplicated in the machine cycle for IN/OUT instruction. **[ESE-EE-10]**

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

10.

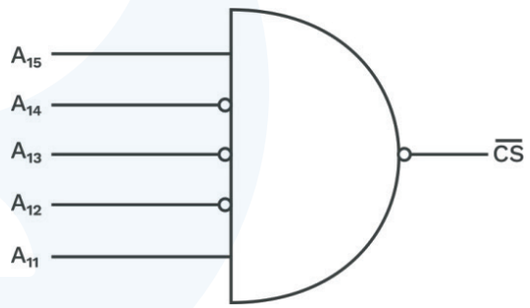


Fig. 3.13

The logic circuit used to generate the active low chip select signal $(\overline{CS})$ by an Intel 8085 microprocessor to address a peripheral is shown in the figure. The peripheral will respond to address in the range: **[ESE-EE-11]**

(A) 7800 H–7FFF H
(B) 100 H–1FFF H
(C) 8000 H–8FFF H
(D) 8800 H–8FFF H

11. In an 8085 microprocessor system, the active low chip select $(\overline{CS})$ signal is generated by passing address lines A_{15}, ..., A_{10} through a six inputs NAND gate. For selecting the address

range CC00 to CFFF, the inputs to the NAND gate are: **[BSNL-JTO-TC-09]**

(A) $A_{10}, A_{11}, \overline{A}_{12}, \overline{A}_{13}, A_{14}, A_{15}$

(B) $\overline{A}_{10}, \overline{A}_{11}, A_{12}, A_{13}, \overline{A}_{14}, \overline{A}_{15}$

(C) $A_{10}, A_{11}, \overline{A}_{12}, A_{13}, A_{14}, A_{15}$

(D) $A_{10}, A_{11}, A_{12}, A_{13}, A_{14}, A_{15}$

12. The range of the address of the RAM, which is interfaced to a microprocessor as shown in the figure is: **[BSNL-JTO-EE-09]**

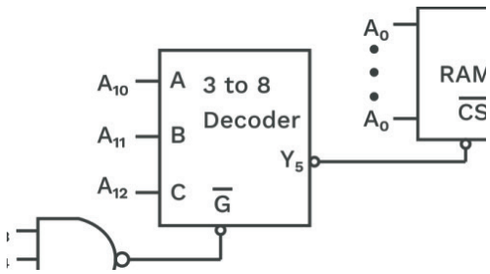


Fig. 3.14

(A) 1400–17FF

(B) E400–EFFF

(C) F000–FFFF

(D) F400–F7FF

Answer Key

Level I

1. (D) **2.** (C) **3.** (B) **4.** (C) **5.** (D) **6.** (B) **7.** (A) **8.** (C) **9.** (B) **10.** (B)
11. (B) **12.** (A) **13.** (B) **14.** (C) **15.** (A)

Level II

1. (A) **2.** (C) **3.** (B) **4.** (C) **5.** (A) **6.** (B) **7.** (B) **8.** (B) **9.** (B) **10.** (D)
11. (A) **12.** (D)